

SHAGUN SINGH

Data Engineer | 4.5+ Years Experience

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PROFESSIONAL SUMMARY

AWS Data Engineer with 4.5+ years building production-grade data pipelines and automation solutions across Banking and Automotive domains. Experienced in Python/Selenium-based ingestion, PySpark transformations on AWS Glue, S3-based data processing, and delivering analytics-ready datasets into Amazon Redshift. Skilled at replacing manual workflows with scalable, auditable automation frameworks that improve data accuracy and operational efficiency.

TECHNICAL SKILLS

AWS Services	S3, AWS Glue (ETL, Crawler, Catalog, Data Quality), Redshift, Athena, Lambda, DynamoDB, Kinesis Streams, EventBridge, SNS
Languages	Python, PySpark, SQL
Automation	Selenium-based workflow automation, dynamic web navigation, form processing, audit reporting
Data Processing	Pandas, OpenPyXL, Apache Spark, Hadoop
Orchestration	Apache Airflow - DAG design, scheduling, failure alerting
Data Stores	Amazon Redshift, Hive, MySQL, DynamoDB
Dev Practices	Git, YAML config management, centralised logging, audit reporting, BRD-to-code delivery

WORK EXPERIENCE

Systems Engineer - *Tata Consultancy Services*

Dec 2021 – Present

Project 1- Cloud Data Pipeline - ETL & Analytics | Client: Banking & Financial Services

Tech Stack: Python, Selenium, Apache Airflow, AWS Glue (PySpark), S3, Redshift, Pandas, YAML, HTML Reporting

- **Architected and owned an end-to-end Airflow-orchestrated ETL pipeline** extracting data from multiple web applications via Python and Selenium, standardising outputs to S3, and delivering analytics-ready datasets into Redshift - eliminating all manual extraction effort and cutting data turnaround from days to hours.
- **Engineered 5+ PySpark transformation jobs on AWS Glue** performing cleansing, deduplication, aggregations, and business-rule transformations at scale - enabling analytics and reporting teams to consume validated Redshift data directly without additional processing.
- **Implemented a multi-layer data reconciliation and quality framework** aligned to BRD requirements, catching discrepancies at source, mid-pipeline, and post-load - achieving near-zero data inconsistencies in a regulated financial services environment.
- **Eliminated hardcoded credentials and environment-specific code** by externalising all pipeline configs, credentials, and transformation parameters into YAML - cutting cross-environment deployment time by ~60% and meeting security compliance requirements.
- **Built automated HTML email alerting triggered post-ETL** notifying stakeholders of data availability, exceptions, and reconciliation failures within minutes of pipeline completion - replacing manual status-update calls entirely.

Project 2 · Pricing Workflow Automation Platform | Client: Automotive

Tech Stack: Python, Selenium, Pandas, OpenPyXL, AWS S3, Logging Framework

- **Eliminated 100% of manual pricing data-entry** for a high-volume enterprise pricing portal by engineering an end-to-end Python + Selenium automation platform - processing bulk option-code and pricing updates that previously required hours of repetitive operator effort.
- **Built a production-grade Selenium framework** capable of handling complex multi-step navigation, dynamic iframe switching, tab transitions, and form interactions across the pricing management portal - zero human intervention required at runtime.
- **Engineered an S3-driven ingestion pipeline** to retrieve multi-sheet Excel files, validate and map option-code/pricing datasets using Pandas and OpenPyXL, and apply configurable business logic - reducing pricing update errors to near-zero.
- **Delivered full audit traceability** by capturing application-level success/error responses per option code, generating consolidated audit-ready reports, and archiving execution logs back to S3 - enabling compliance review and downstream operational reporting without manual record-keeping.

INDEPENDENT PROJECTS

Gadget Sales Data Projection ([View my Project](#))

- Built a real-time Change Data Capture pipeline: DynamoDB, DynamoDB Streams, EventBridge Pipes, Kinesis Streams, Firehose, S3, with Athena for ad-hoc querying - reducing data latency from daily batch to sub-minute streaming.
- Enabled real-time sales dashboards and instant anomaly detection by streaming 100% of sales record changes through the pipeline with no data loss.

Quality Movie Data Analysis ([View my Project](#))

- Engineered a quality-gated pipeline using AWS Glue Data Quality checks to reject non-conforming records before Redshift load, ensuring only validated data enters the warehouse - zero silent data failures in production.
- Implemented EventBridge - SNS alerting with automatic quarantine routing of failed records to a separate S3 bucket, giving data ops teams full visibility and a clear remediation path.

EDUCATION

B.Tech - Computer Science Engineering - IMS Engineering College, AKTU, Uttar Pradesh

Aug 2017 - Aug 2021, Grade - 78.8% (First Class with HONS)

ACHIEVEMENTS

- **"On The Spot" Award - TCS FY23:** Recognised for outstanding delivery on a high-visibility client data engineering engagement.
- **SQL Gold Badge - HackerRank:** SQL proficiency validated by external assessment.